



[Srivijayesh Venugopal]

Continuous Analog Wave Computation in Porous Media

Faculty of Engineering and Applied Sciences
[Computational Fluid Dynamics]

MSc
Academic Year: 2025–2026

Supervisor: [Dr Supervisor Name]
September 2026



Faculty of Engineering and Applied Sciences
[Computational Fluid Dynamics]

MSc

Academic Year: 2025–2026

[Srivijayesh Venugopal]

Continuous Analog Wave Computation in Porous Media

Supervisor: [Dr Supervisor Name]
September 2026

This thesis is submitted in partial fulfilment of the
requirements for the degree of MSc.

© Cranfield University 2026. All rights reserved. No part of
this publication may be reproduced without the written
permission of the copyright owner.

Academic Integrity Declaration

I declare that:

- the thesis submitted has been written by me alone.
- the thesis submitted has not been previously submitted to this university or any other.
- that all content, including primary and/or secondary data, is true to the best of my knowledge.
- that all quotations and references have been duly acknowledged according to the requirements of academic research.

I understand that to knowingly submit work in violation of the above statement will be considered by examiners as academic misconduct.

Chapter 1

Abstract

[Write abstract here (not more than 300 words.)]

Keywords:

physical neural networks, acoustic wave computing, inverse design, Johnson–Champoux–Allard model, Helmholtz equation, porous media, equivalent-fluid models, neuromorphic engineering

Chapter 2

Acknowledgements

[Optional: acknowledge support, supervisors, funding bodies, etc.]

Contents

Academic Integrity Declaration	1
1 Abstract	2
2 Acknowledgements	3
Contents	4
List of Figures	6
List of Tables	7
3 List of Abbreviations	8
4 Introduction	1
4.1 Background and Motivation	1
4.2 Research Aim	2
4.3 Research Objectives	2
4.4 Scope and Delimitations	2
4.5 Thesis Structure	3
5 Literature Review	4
5.1 Introduction	4
5.2 Physical Neural Networks	4
5.3 Acoustic Wave Computing	5
5.4 Porous Media Acoustics	6
5.4.1 Equivalent-Fluid Models	6
5.4.2 Porous Materials in Computing Contexts	6
5.4.3 Metaporous Materials	7
5.5 Inverse Design and Optimisation	7
5.6 Synthesis and Research Gap	8
6 Methodology	10
6.1 Overview	10
6.2 Computational Domain	10
6.3 Governing Equations	11
6.3.1 Linear Acoustic Helmholtz Equation	11
6.3.2 Johnson-Champoux-Allard Effective Properties	12

6.3.3	Zwikker-Kosten Simplification	12
6.3.4	Complex Wavenumber and Characteristic Impedance	13
6.4	Numerical Discretisation	13
6.4.1	Spatial Discretisation and Boundary Conditions	13
6.4.2	Frequency-Domain Solver	13
6.4.3	Source Injection and Probe Readout	14
6.5	Design Parameterisation	14
6.5.1	Block-Based Design Variables	14
6.5.2	Smooth Mapping to the Simulation Grid	14
6.5.3	Fixed Geometric Parameters	15
6.6	Optimisation Framework	15
6.6.1	Wave-Computing Task	15
6.6.2	Objective Function	15
6.6.3	Parameter Bounds and Transforms	16
6.6.4	Optimisation Algorithm	16
7	Results and Discussion	17
7.1	Overview	17
7.2	Helmholtz Solver Validation	17
7.2.1	Free-Air Propagation	17
7.2.2	Hard-Wall Cavity Modes	17
7.2.3	Uniform Porous Slab	17
7.3	Design-Region Demonstrations	17
7.3.1	Task Definition	17
7.3.2	Baseline Designs	17
7.3.3	Optimised Designs	17
7.4	Design-Space Ablations	18
7.5	Discussion	18
8	Conclusions and Future Work	19
8.1	Summary of Contributions	19
8.2	Limitations	19
8.3	Future Work	19
8.4	Concluding Remarks	19
	References	20
A	Additional Derivations	23

List of Figures

6.1	Simulation-first optimisation pipeline.	10
6.2	Computational domain sketch for the duct configuration. The left and right ends are absorbing PMLs; the top and bottom boundaries are hard walls. The design region sits between the source and the probe array. . . .	11

List of Tables

5.1	Comparison of the present work with the closest related approaches. . . .	9
6.1	Symbols used in the Johnson-Champoux-Allard model.	12

Chapter 3

List of Abbreviations

FDTD	Finite-Difference Time-Domain
CFD	Computational Fluid Dynamics
BC	Boundary Condition
PML	Perfectly Matched Layer
CFL	Courant–Friedrichs–Lewy
PDE	Partial Differential Equation
ODE	Ordinary Differential Equation
RNN	Recurrent Neural Network
PNN	Physical Neural Network
ZK	Zwikker–Kosten
JCA	Johnson–Champoux–Allard
TMM	Transfer-Matrix Method
MOF	Metal–Organic Framework
PINN	Physics-Informed Neural Network

Chapter 4

Introduction

4.1 Background and Motivation

Artificial neural networks have become the dominant paradigm for tasks such as image classification, speech recognition, and scientific prediction, yet their deployment on conventional digital hardware faces growing constraints. Inference is discretised into clocked arithmetic operations, memory is physically separated from processing units, and high-throughput applications demand ever-larger allocations of power, chip area, and cooling. The resulting energy consumption, latency, and hardware cost motivate the search for alternative computational substrates that can perform inference directly in the physical domain, sidestepping the von Neumann bottleneck by allowing data to be processed “in flight” rather than shuttled between processor and memory.

The original motivation for this project was a broader class of *fluid neural networks*, in which reaction, diffusion, and flow in a fluid medium would implement continuous, parallel computation. While chemically reacting flows offer a rich physical substrate, a tractable forward model that is both faithful to the chemistry and suitable for inverse design was not available within the MSc timeframe. This led to a more focused investigation of acoustic wave propagation in a structured fluid-saturated medium, where the governing physics is well described by the Helmholtz equation and the design space can be parameterised by measurable material properties. Chapter 5 reviews the wider field of physical neural networks and acoustic wave computing that motivates this choice.

This project investigates a specific class of acoustic physical neural network: a continuous analog wave computer implemented in a fluid-saturated porous medium. The substrate is a rigid-frame porous solid through which acoustic waves propagate; the pore architecture modulates the effective acoustic density and bulk modulus of the air inside the pores, and the spatial distribution of porosity, flow resistivity, and tortuosity therefore acts as the trainable “weight” field of the network. Compared with free-space or layered acoustic computers, a porous medium offers a compact, three-dimensional, manufacturable volume in which wave interference and viscous damping can be exploited for computation. An earlier exploration considered using temperature fields in air to modulate sound speed, but strong frequency selectivity required impractically large devices. The porous-medium direction was chosen because it offers stronger scattering and damping per wavelength, together with a direct set of trainable material parameters. Despite the maturity of equivalent-fluid porous models [1], the use of spatially varying porous

properties for analog inference remains largely unexplored.

4.2 Research Aim

The aim of this project is to investigate whether a rigid-frame fluid-saturated porous medium can be designed, by spatial variation of its equivalent-fluid properties, to perform simple analog wave-computing tasks in the speech-frequency band.

4.3 Research Objectives

The specific objectives are:

1. To formulate the acoustic wave equation in a rigid-frame porous medium using the Johnson-Champoux-Allard (JCA) equivalent-fluid model in the frequency domain.
2. To develop a Helmholtz solver that accepts spatially varying JCA parameters and reproduces analytical wave-propagation behaviour in free air and in uniform porous slabs.
3. To parameterise a design region by coarse blocks of porosity, flow resistivity, and high-frequency tortuosity, mapped smoothly onto the simulation grid.
4. To optimise the design-region parameters for a simple wave-computing task and compare design-space reductions (e.g. ϕ -only, $\phi + \sigma$, and $\phi + \sigma + \alpha_\infty$).

4.4 Scope and Delimitations

The project is deliberately scoped to remain feasible within an MSc timeframe while still addressing a research gap. The following aspects are in scope:

- Two-dimensional, time-harmonic acoustic wave propagation in a rigid-frame porous medium.
- The Johnson-Champoux-Allard equivalent-fluid model with spatially varying porosity, flow resistivity, and high-frequency tortuosity.
- Frequency-domain Helmholtz solves over the speech-frequency band (approximately 300 Hz to 4 kHz).
- A block-wise design parameterisation mapped smoothly onto the simulation grid.
- Gradient-free optimisation of the block parameters for a simple wave-computing task.

The following aspects are outside the present scope:

- Full poroelastic Biot theory; the porous frame is treated as rigid and motionless.

- Direct pore-scale simulation; pores are modelled only through their effective macroscopic properties.
- Time-domain transient effects; the formulation assumes linear, time-invariant steady-state response.
- Nonlinear activation functions and recurrent time-domain dynamics; the present demonstration is a linear, frequency-domain processor.
- Coupled microstructural modelling of the characteristic lengths Λ and Λ' ; these are fixed to representative values rather than derived from the block-wise porosity, resistivity, and tortuosity.
- Adjoint-based or differentiable optimisation; the design space is small enough for gradient-free search.

4.5 Thesis Structure

The remainder of this thesis is organised as follows. Chapter 5 reviews physical neural networks, acoustic analog computing, equivalent-fluid porous-media models, and inverse design methods. Chapter 6 presents the JCA Helmholtz formulation, the block-based design parameterisation, the optimisation framework, and the validation strategy. Chapter 7 reports solver validation, design-space reductions, and the optimised wave-computing demonstrations. Chapter 8 summarises the contributions, limitations, and future directions.

Chapter 5

Literature Review

5.1 Introduction

The remarkable success of artificial neural networks in tasks such as image classification, natural language processing, and scientific prediction has been built on digital hardware that discretises computation into clocked arithmetic operations. As models grow in size and inference is pushed toward edge devices, the energy cost, latency, and chip area of this approach have motivated a search for alternative computational substrates in which the physics of the device itself performs useful transformations. This chapter reviews the literature on such physical neural networks, with particular attention to wave-based systems and to the porous acoustic medium proposed in this thesis. The review begins with the broad idea of physical computation, narrows to acoustic wave computers, then to equivalent-fluid porous acoustics, and concludes with inverse-design methods and the specific research gap addressed by this project.

5.2 Physical Neural Networks

A physical neural network is a material or device whose internal physical dynamics implement the mathematical operations of a neural network. Rather than simulating those dynamics in software, the network is the physical system itself: inputs are encoded as fields or excitations, the material transforms them through its natural evolution, and outputs are read from probes or detectors. The appeal of this approach is that computation can occur in parallel across space and continuously in time, without the repeated memory transfers that dominate digital inference.

Optical systems have been among the most prominent demonstrations. Lin et al showed that a stack of diffractive surfaces can be trained to perform image classification and other tasks entirely through light propagation [2]. The layers modulate the phase of the optical field, and interference in free space produces the desired mapping. This work established that a physical wave system can be designed as a deep neural network, with the geometry of the diffractive elements serving as the trainable weights.

Mechanical systems provide a complementary view. Stern et al demonstrated supervised learning through physical changes in a mechanical network of springs and masses [3], and later developed a broader theoretical framework for learning in physical net-

works [4]. In these systems the equilibrium or linear response of the network encodes the learned function, and training adjusts physical parameters such as spring constants rather than digital weights.

More recently, Wright et al extended the physical neural network idea to deep architectures by chaining multiple physical systems and training them with backpropagation through differentiable simulations [5]. Momeni et al addressed a key practical difficulty of that approach, namely that real physical systems are noisy and difficult to differentiate through, by proposing a backpropagation-free training method for deep physical neural networks [6]. Together these works show that physical networks can be both deep and trainable, but they also highlight the challenge of transferring designs from simulation to physical hardware.

A unifying theoretical perspective was provided by Hughes et al, who showed that linear wave propagation can be written as an analog recurrent neural network [7]. In this view, the state of the wave field at each time step is the hidden state of the RNN, and the evolution operator is a sparse linear recurrence whose coefficients depend on the local material properties. This framework is relevant to the present work because it treats the acoustic field as the computational state and the material distribution as the trainable parameters. The present work does not implement the time-domain recurrent formulation directly; instead, it adopts a frequency-domain Helmholtz formulation as a tractable first step toward trainable acoustic processors of this kind.

5.3 Acoustic Wave Computing

Acoustic waves are an attractive substrate for physical computation because they propagate in ordinary fluids and solids, interact strongly with structured boundaries, and can be generated and detected with simple transducers. Early work in acoustic metamaterials showed that subwavelength scatterers can be arranged to perform mathematical operations on incident wave fields. Silva et al demonstrated that metamaterial slabs could implement spatial differentiation, integration, and other analog mathematical operations [8], and Zangeneh-Nejad et al provided a broader review of analog computing with metamaterials [9].

Another line of research has pursued acoustic logic gates and routing. Zuo et al developed an acoustic analog computing system based on labyrinthine metasurfaces, in which phase manipulation in structured channels implements Boolean operations [10]. Zhang et al used self-collimating acoustic beams in phononic crystals to construct AND, OR, NOT, and XOR gates [11], and later extended the idea to compact transformable logic gates based on density-near-zero metamaterials [12]. Wang et al proposed binary-phase acoustic passive logic gates [13], and Xia et al demonstrated an ultra-broadband acoustic logic gate using passive phase manipulation [14]. Parsa et al used evolutionary search to embed AND and XOR gates in two-dimensional granular metamaterials by arranging mass- or stiffness-contrasting particles [15]. These papers establish that acoustic interference and scattering can produce useful computational primitives, but they rely on carefully engineered free-space, channel, or granular geometries rather than on continuously varying material properties.

The present work differs from these acoustic logic-gate studies in two respects. First, the medium is a porous solid rather than a free-space or channel geometry, which changes

the governing physics from ordinary acoustics to equivalent-fluid porous acoustics. Second, the design space is parameterised by continuous fields of porosity, flow resistivity, and tortuosity, allowing gradient-free optimisation to discover structures that would be difficult to design by hand.

A final consideration is whether to formulate the problem in the time domain or the frequency domain. Hughes et al’s RNN framework is naturally time-domain: the acoustic state evolves through discrete time steps, and the system can be trained like a recurrent neural network [7]. Time-domain finite-difference methods are well developed for room acoustics and musical acoustics [16, 17, 18, 19]. However, the Johnson-Champoux-Allard model of porous acoustics is most naturally expressed in the frequency domain, because the effective density and bulk modulus are frequency-dependent complex quantities. For this reason the present work adopts a frequency-domain Helmholtz formulation, solving for each frequency independently and reconstructing time-domain signals through superposition.

5.4 Porous Media Acoustics

5.4.1 Equivalent-Fluid Models

The acoustics of fluid-saturated porous media can be approached at two levels. The more general theory, due to Biot, treats both the fluid and the solid frame as elastic continuums coupled at the pore boundaries [1]. Biot’s theory predicts two compressional waves and one shear wave in the porous medium, and it remains the foundation for poroelastic analysis. For many air-saturated materials with a heavy, stiff frame, however, the frame can be treated as rigid and motionless, and the air in the pores behaves as an equivalent fluid with frequency-dependent effective properties. This equivalent-fluid approximation greatly simplifies the governing equations and is accurate when the acoustic wavelength is much larger than the pore size.

The earliest equivalent-fluid model for sound absorbing materials was developed by Zwikker and Kosten [20]. Their model accounts for viscous losses in the pores through a complex effective density and forms the basis for later, more detailed descriptions. The Johnson-Champoux-Allard (JCA) model extends this approach by including both dynamic tortuosity and dynamic bulk modulus, with separate viscous and thermal characteristic lengths [21, 22]. The full theory and its applications are described in the reference text by Allard and Atalla [23]. The JCA model is used in the present work because it captures the frequency-dependent damping and dispersion of sound in rigid-frame porous materials with a small number of measurable parameters.

Nonlinear extensions of porous acoustics exist, including Biot’s own nonlinear theory of porous solids [24] and more recent work on nonlinear wave propagation in porous materials. These are not considered here because the speech-band amplitudes of interest are small enough that linear acoustics is an excellent approximation.

5.4.2 Porous Materials in Computing Contexts

Porous materials have been used in several computing-related contexts, though rarely as continuous acoustic inference substrates. Jaafar et al demonstrated mesoporous silica

memristors for neuromorphic switching and reservoir computing [25], and Ding et al reviewed porous crystalline materials for memories and neuromorphic computing systems [26]. These works exploit the pore structure for electronic or ionic switching rather than for acoustic wave interference. They nevertheless support the broader motivation that porous media are a promising, manufacturable platform for unconventional computing.

5.4.3 Metaporous Materials

The term *metaporous* refers to engineered porous structures that combine a porous substrate with resonant or structured elements to obtain acoustic behaviour beyond that of ordinary porous materials. Typical examples include perforated or micro-perforated face plates backed by porous layers, porous matrices with embedded Helmholtz resonators, and periodic three-dimensional lattice cells. These structures are designed at the unit-cell or microstructure level and are usually analysed with finite-element or acoustic-structural models rather than with an equivalent-fluid approximation. Metaporous absorbers can achieve broadband or deep-subwavelength absorption by tuning the geometry of the resonant features, and inverse-design studies such as Pan et al have optimised metaporous topologies for target absorption spectra. The present work differs from metaporous design because it treats the porous medium as a continuous equivalent fluid described by the JCA model. The design variables are macroscopic effective properties rather than unit-cell geometry, and the objective is wave-based computation rather than absorption.

Other physical computing substrates provide useful context. Adamatzky reviewed the history of liquid computers, in which fluids or chemical reactions perform computation [27]. Baltussen et al demonstrated chemical reservoir computation in a self-organizing reaction network [28], and Prakash and Gershenfeld developed microfluidic bubble logic, in which gas bubbles in channels carry and process information [29]. These systems are not acoustic, but they illustrate a long tradition of using physical and chemical media for computation. The gap addressed by this thesis is the use of porous acoustic media specifically as a trainable wave-computing substrate.

5.5 Inverse Design and Optimisation

Designing a wave-computing device by hand becomes difficult as the number of design parameters grows, so inverse design methods are essential. Topology optimisation is a well-established approach in which a design field is allowed to vary continuously across the domain and is driven toward an optimal configuration by gradients of an objective function [9]. Density-based methods use a filtered and projected material distribution to obtain manufacturable binary or multi-material designs. These methods are powerful but typically require adjoint gradients and careful regularisation to avoid small-scale features.

Inverse design has also been applied to conventional porous and metaporous materials, although with objectives quite different from wave computing. Boulvert et al optimised one-dimensional through-thickness gradients of Johnson-Champoux-Allard-Lafarge parameters to achieve broadband perfect absorption [30]. Pan et al used a convolutional-neural-network surrogate coupled to a genetic algorithm to inversely design two-dimensional metaporous geometries for target sound-absorption spectra [31]. These studies demon-

strate that spatially varying porous parameters can be optimised, but they do so for absorption or transmission rather than for analog computation.

When gradients are unavailable or unreliable, gradient-free optimisation becomes attractive. This is the case in the present work, where the design variables are coarse blocks of porous parameters and the objective function is evaluated by solving the Helmholtz equation. Gradient-free methods such as differential evolution, covariance matrix adaptation, or Bayesian optimisation can explore small design spaces without requiring derivatives. They scale poorly to very large numbers of variables, but for a block-wise parameterisation with tens or hundreds of variables they are practical and robust.

Adjoint-based and differentiable solvers represent the state of the art for larger design spaces. In the frequency-domain Helmholtz setting, the adjoint method allows the gradient of an objective function with respect to many design parameters to be computed at the cost of one additional solve. Physics-informed neural networks and automatic differentiation provide alternative routes to gradient-based design. These approaches are not used in the present work, but they define the direction in which future work could scale the design space.

5.6 Synthesis and Research Gap

The literature reviewed above establishes three important points. First, physical neural networks are a viable alternative to digital inference, with demonstrations in optics, mechanics, and acoustics. Second, acoustic waves can be used to implement computational primitives such as logic gates and mathematical operations through interference and scattering. Third, porous media have well-developed equivalent-fluid models and are increasingly used in neuromorphic and unconventional computing contexts.

What is missing is an inverse-designed two-dimensional rigid-frame porous acoustic processor in which the medium is parameterised by continuous effective-fluid properties and optimised for a wave-computing task such as frequency classification or routing. Existing acoustic computers rely on free-space, channel, or granular geometries, while existing porous-media inverse design focuses on graded absorption layers or metaporous unit-cell topologies for absorption. This project addresses that gap by formulating the JCA Helmholtz equation in a design region, parameterising the medium by blocks of porosity, flow resistivity, and tortuosity, and optimising those blocks for a simple wave-computing task.

Table 5.1: Comparison of the present work with the closest related approaches.

	Boulvert et al 2019	Pan et al 2023	Parsa et al 2022	This work
Substrate	Graded porous layer	2D metaporous structure	Granular metamaterial	Rigid-frame porous equivalent fluid
Dimension	1D through-thickness	2D unit-cell geometry	2D discrete particles	2D continuous effective medium
Design variables	JCAL parameters vs. depth	Unit-cell topology	Particle positions	$\phi, \sigma, \alpha_\infty$ blocks
Objective	Broadband absorption	Target absorption spectra	AND / XOR logic gates	Wave-computing task

Chapter 6

Methodology

6.1 Overview

The methodology follows a simulation-first pipeline. A two-dimensional acoustic source injects time-harmonic waves into a domain that contains a design region. The design region is parameterised by coarse blocks of porosity, flow resistivity, and high-frequency tortuosity, which are mapped smoothly onto the simulation grid. For each frequency of interest, the Helmholtz equation is solved with spatially varying Johnson-Champoux-Allard (JCA) effective properties. Probe pressures are extracted, and a gradient-free optimiser adjusts the block values to minimise a task-specific objective. The resulting designs are validated against analytical transfer-matrix predictions for uniform porous slabs. The overall workflow is shown in Fig. 6.1.

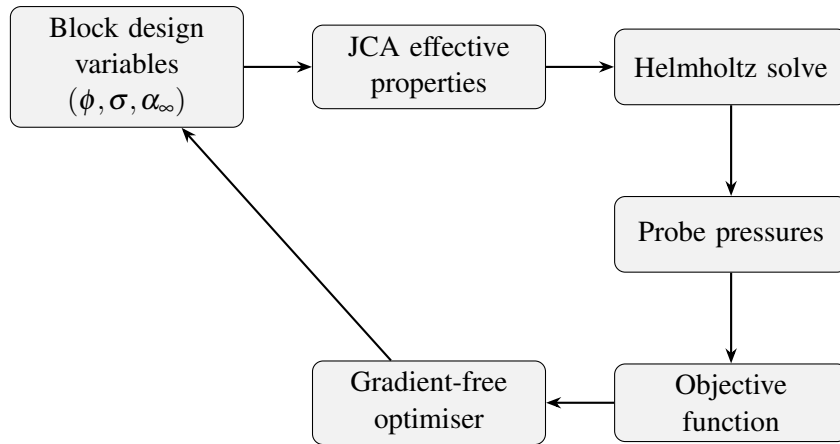


Figure 6.1: Simulation-first optimisation pipeline.

6.2 Computational Domain

The computational domain is a bounded, connected region $\Omega \subset \mathbb{R}^2$ with boundary $\partial\Omega$. The boundary is partitioned into two disjoint subsets:

- Γ_{abs} : the absorbing boundary, where a PML mimics free-field radiation;

- Γ_{wall} : the hard-wall boundary, where the normal particle velocity vanishes.

Thus $\partial\Omega = \Gamma_{\text{abs}} \cup \Gamma_{\text{wall}}$ and the two subsets intersect only at corners. In a duct configuration the left and right ends are absorbing, while the top and bottom are hard walls. In a free-field configuration all four sides are absorbing. The duct configuration used in this work is illustrated in Fig. 6.2.

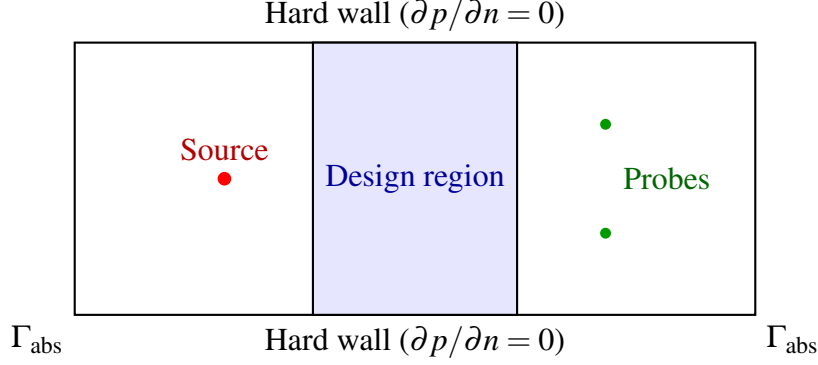


Figure 6.2: Computational domain sketch for the duct configuration. The left and right ends are absorbing PMLs; the top and bottom boundaries are hard walls. The design region sits between the source and the probe array.

Inside the domain lies the design region $\Omega_{\text{design}} \subset \Omega$, a rectangular sub-domain in which the porous material properties may vary. The complement $\Omega \setminus \Omega_{\text{design}}$ is filled with air, so that

$$\phi(\mathbf{x}) = 1, \quad \sigma(\mathbf{x}) = 0, \quad \alpha_{\infty}(\mathbf{x}) = 1 \quad \text{for } \mathbf{x} \in \Omega \setminus \Omega_{\text{design}}. \quad (6.1)$$

Within Ω_{design} the fields are parameterised by coarse blocks as described in Sec. 6.5.

The source is located a short distance inside the domain, at position $\mathbf{x}_s \in \Omega$, and is represented as a source term in the Helmholtz equation. A set of N_p probe locations $\{\mathbf{x}_p\}_{p=1}^{N_p}$ is fixed in Ω . These probes record the complex pressure $p(\omega, \mathbf{x}_p)$ for each frequency. The source amplitude $S(\omega)$ is normalised so that the transfer function $H(\omega, \mathbf{x}_p) = p(\omega, \mathbf{x}_p)/S(\omega)$ is independent of source strength.

6.3 Governing Equations

6.3.1 Linear Acoustic Helmholtz Equation

For time-harmonic acoustic pressure $p(\mathbf{x})$ at angular frequency ω , the Helmholtz equation in an inhomogeneous effective fluid is

$$\nabla \cdot \left(\frac{1}{\tilde{\rho}(\omega, \mathbf{x})} \nabla p \right) + \frac{\omega^2}{\tilde{K}(\omega, \mathbf{x})} p = S(\omega) \delta(\mathbf{x} - \mathbf{x}_s), \quad (6.2)$$

where $\tilde{\rho}$ is the complex effective density, $\tilde{K}$ is the complex effective bulk modulus, and the right-hand side represents a point source at $\mathbf{x}_s$ with frequency-domain amplitude $S(\omega)$. When the coefficients vary with position they must remain inside the divergence operator; pulling them outside would neglect the gradient terms that arise at material interfaces.

6.3.2 Johnson-Champoux-Allard Effective Properties

The JCA model treats the air in a rigid-frame porous medium as an equivalent fluid with frequency-dependent complex properties. The effective density is

$$\tilde{\rho}(\omega, \mathbf{x}) = \frac{\rho_0 \alpha_\infty(\mathbf{x})}{\phi(\mathbf{x})} \left[1 + \frac{\sigma(\mathbf{x}) \phi(\mathbf{x})}{j \omega \rho_0 \alpha_\infty(\mathbf{x})} \sqrt{1 + j \frac{4 \alpha_\infty(\mathbf{x})^2 \eta \rho_0 \omega}{\sigma(\mathbf{x})^2 \phi(\mathbf{x})^2 \Lambda(\mathbf{x})^2}} \right], \quad (6.3)$$

and the effective bulk modulus is

$$\tilde{K}(\omega, \mathbf{x}) = \frac{\gamma P_0}{\phi(\mathbf{x})} \left[\gamma - \frac{\gamma - 1}{1 + \frac{8 \eta}{j \omega \rho_0 Pr \Lambda'(\mathbf{x})^2} \sqrt{1 + j \frac{\omega \rho_0 Pr \Lambda'(\mathbf{x})^2}{16 \eta}}} \right]^{-1}. \quad (6.4)$$

The symbols in these equations are listed in Tab. 6.1. The model assumes a rigid, motionless frame and cylindrical pore geometry. It is valid when the acoustic wavelength is much larger than the pore size, which holds comfortably for speech-band frequencies in air and pore scales of tens of micrometres.

Table 6.1: Symbols used in the Johnson-Champoux-Allard model.

Symbol	Meaning	Units
ρ_0	Ambient air density	kg m^{-3}
η	Dynamic viscosity of air	Pa s
γ	Heat capacity ratio of air	—
P_0	Ambient pressure	Pa
Pr	Prandtl number of air	—
ϕ	Porosity	—
σ	Static flow resistivity	Pa s m^{-2}
α_∞	High-frequency tortuosity	—
Λ	Viscous characteristic length	m
Λ'	Thermal characteristic length	m

6.3.3 Zwikker-Kosten Simplification

The Zwikker-Kosten (ZK) model is the low-frequency limit of the JCA model, obtained when the viscous penetration depth is much larger than the pore size. In this limit the effective density and bulk modulus become real and frequency independent:

$$\rho_{\text{eff}}(\mathbf{x}) = \frac{\rho_0 \alpha_\infty(\mathbf{x})}{\phi(\mathbf{x})}, \quad (6.5)$$

$$K_{\text{eff}}(\mathbf{x}) = \frac{\rho_0 c_0^2}{\phi(\mathbf{x})}, \quad (6.6)$$

where c_0 is the ambient sound speed. The corresponding time-domain equations are

$$\rho_{\text{eff}} \frac{\partial \mathbf{u}}{\partial t} + \boldsymbol{\sigma} \mathbf{u} = -\nabla p, \quad (6.7)$$

$$\frac{1}{K_{\text{eff}}} \frac{\partial p}{\partial t} = -\nabla \cdot \mathbf{u}. \quad (6.8)$$

The ZK model is useful as a computationally inexpensive design tool because its time-domain finite-difference formulation avoids assembling and solving a complex sparse linear system at each frequency. It does not capture the frequency-dependent viscous and thermal losses of the full JCA model, and its validity should be checked against the JCA model for any design that is carried forward.

6.3.4 Complex Wavenumber and Characteristic Impedance

From the effective density and bulk modulus, the complex wavenumber and characteristic impedance are

$$k(\omega, \mathbf{x}) = \omega \sqrt{\frac{\tilde{\rho}(\omega, \mathbf{x})}{\tilde{K}(\omega, \mathbf{x})}}, \quad (6.9)$$

$$Z_c(\omega, \mathbf{x}) = \sqrt{\tilde{\rho}(\omega, \mathbf{x}) \tilde{K}(\omega, \mathbf{x})}. \quad (6.10)$$

These quantities are used when comparing numerical results against analytical transfer-matrix predictions for uniform porous slabs.

6.4 Numerical Discretisation

6.4.1 Spatial Discretisation and Boundary Conditions

The computational domain is a rectangular region in two dimensions, discretised on a structured Cartesian grid. The grid spacing is chosen to resolve the shortest wavelength of interest. At 4 kHz the acoustic wavelength in air is approximately 86 mm; a grid spacing of 2 mm gives more than forty points per wavelength, which is sufficient for the second-order finite-difference approximation used here.

On the absorbing boundary Γ_{abs} a Robin-type Absorbing Boundary condition (might change to a perfectly matched layer) is used to mimic free-field radiation [32]. The layer has a width of approximately one wavelength at the highest frequency of interest. On the hard-wall boundary Γ_{wall} the normal particle velocity vanishes, which in pressure form is the Neumann condition $\partial p / \partial n = 0$.

6.4.2 Frequency-Domain Solver

For each frequency ω the Helmholtz equation (6.2) is assembled into a sparse linear system. The spatially varying coefficients $\tilde{\rho}(\omega, \mathbf{x})$ and $\tilde{K}(\omega, \mathbf{x})$ are evaluated at each grid point using the local block-interpolated values of ϕ , σ , and α_∞ . The system is solved with a direct sparse solver for small to moderate grid sizes, or with an iterative solver for larger problems. Direct solvers are preferred here because the number of frequency samples is modest and each solve must be reliable during optimisation.

6.4.3 Source Injection and Probe Readout

The source is a point source located inside the domain, represented by the delta-function term in (6.2). For broadband signals the Helmholtz equation is solved at a set of discrete frequencies that span the signal bandwidth, and the probe responses are combined by inverse Fourier transform or convolution with the input spectrum. Probes record the complex pressure $p(\omega, \mathbf{x}_{\text{probe}})$ at fixed locations. The time-domain probe signal $p(t, \mathbf{x}_{\text{probe}})$ for an input $s(t)$ is obtained from

$$p(t, \mathbf{x}_{\text{probe}}) = \mathcal{F}^{-1} \{ H(\omega, \mathbf{x}_{\text{probe}}) S(\omega) \}, \quad (6.11)$$

where $S(\omega)$ is the Fourier transform of the input, $H(\omega, \mathbf{x}_{\text{probe}}) = p(\omega, \mathbf{x}_{\text{probe}})/p_{\text{source}}(\omega)$ is the transfer function at the probe, and $\mathcal{F}^{-1}$ denotes the inverse Fourier transform. This frequency-domain superposition is valid because the JCA medium is linear and time-invariant.

6.5 Design Parameterisation

6.5.1 Block-Based Design Variables

The design region is divided into $N_x \times N_y$ coarse rectangular blocks. Each block b carries three design variables whose values are uniform within the block:

- porosity $\phi_b \in [0.30, 0.95]$,
- static flow resistivity $\sigma_b \in [0, 10^5] \text{ Pa s m}^{-2}$,
- high-frequency tortuosity $\alpha_{\infty, b} \in [1.0, 2.5]$.

The bounds are chosen to keep the JCA equivalent-fluid model physically valid. The lower porosity bound ensures that the air-filled pore space remains connected enough for the equivalent-fluid assumption to hold, while the upper bound prevents the limit of pure air. The resistivity range spans from effectively lossless air to highly resistive porous materials typical of dense acoustic foams. The tortuosity range covers straight pores ($\alpha_{\infty} = 1$) to moderately tortuous pore networks.

6.5.2 Smooth Mapping to the Simulation Grid

Block-centred values are mapped onto the fine simulation grid by bilinear interpolation. Each fine-grid point receives a weighted average of the four nearest block values, so jumps between adjacent blocks become smooth ramps on the simulation grid. This parameterisation has three advantages. First, it reduces the number of optimisation variables from thousands of grid points to tens or hundreds of blocks. Second, it suppresses the small-scale checkerboard patterns that often appear in unrestricted topology optimisation. Third, it produces designs that are easier to interpret as a manufacturable arrangement of porous blocks.

6.5.3 Fixed Geometric Parameters

The viscous and thermal characteristic lengths are fixed throughout the domain at

$$\Lambda = 30 \mu\text{m}, \quad \Lambda' = 60 \mu\text{m}, \quad (6.12)$$

corresponding to a target pore scale of approximately $30 \mu\text{m}$. In reality the viscous and thermal characteristic lengths are not independent of porosity and tortuosity; they are set by the microstructure. Fixing them here reduces the design space and avoids non-physical combinations of the five JCA parameters. The values are representative of sintered metals, foams, or 3D-printed lattices with a controllable pore size. A manufacturable interpretation of an optimised design therefore requires selecting a family of microstructures that realises the optimised $(\phi, \sigma, \alpha_\infty)$ at the fixed characteristic lengths. This is a deliberate simplification that treats the medium as an effective-fluid design space rather than as a family of real microstructures, and it limits the physical realism of the resulting designs.

6.6 Optimisation Framework

6.6.1 Wave-Computing Task

The demonstration task used in this work is a frequency binary classifier. Two pure-tone inputs, at frequencies f_1 and f_2 , are injected from the same source location. The optimised medium should route acoustic energy so that one frequency produces a larger response at probe A and the other produces a larger response at probe B. The readout is therefore the complex pressure amplitude at two fixed probe locations. This task is chosen because it requires the medium to exploit frequency-dependent interference and damping, while remaining simple enough for a gradient-free optimiser to converge. Other tasks, such as spatial source classification or logic gates, can be addressed with the same forward model by changing only the objective function and the probe configuration.

6.6.2 Objective Function

For the frequency binary classifier the objective is the mean-squared error between the normalised probe energies and a target distribution. Let $P_A(\omega)$ and $P_B(\omega)$ be the complex pressures at probes A and B for input frequency ω , and let $E_A = |P_A|^2$ and $E_B = |P_B|^2$ be the corresponding energies. The total energy is $E_{\text{tot}} = E_A + E_B$. The target for frequency f_1 is that most of the energy arrives at probe A, while for f_2 most should arrive at probe B. The objective for a single frequency is

$$\mathcal{L}(\omega) = \left(\frac{E_A}{E_{\text{tot}}} - t_A(\omega) \right)^2 + \left(\frac{E_B}{E_{\text{tot}}} - t_B(\omega) \right)^2, \quad (6.13)$$

where $t_A(\omega)$ and $t_B(\omega)$ are the target energy fractions. The total objective is the sum over the two frequencies. The same forward model and solver are used for both frequencies; only the right-hand side source term changes.

6.6.3 Parameter Bounds and Transforms

During optimisation the block variables are kept within their bounds. One approach is to apply smooth parameter transforms, for example

$$\phi_b = 0.30 + \frac{0.65}{1 + \exp(-u_b)}, \quad (6.14)$$

$$\sigma_b = \sigma_{\min} \exp(v_b), \quad (6.15)$$

$$\alpha_{\infty,b} = 1.0 + \text{softplus}(w_b), \quad (6.16)$$

where u_b , v_b , and w_b are the unconstrained optimisation variables. Alternatively, box constraints can be passed directly to the optimiser. The transform approach is more robust against boundary violations during line searches.

6.6.4 Optimisation Algorithm

Because the forward solver is not differentiated in this work, optimisation is performed with a gradient-free method. For the block-wise parameterisation considered here, with tens to hundreds of variables, differential evolution is the chosen algorithm. The objective landscape for porous block parameters is expected to be multimodal because small changes in geometry can create or destroy resonant paths, and differential evolution handles this better than local methods such as Nelder-Mead. The population size and number of generations are chosen to give the optimiser enough evaluations to converge while remaining within the computational budget. The run terminates when the objective function reaches a plateau or the budget is exhausted.

Comparing the final losses and optimised patterns answers whether adding flow resistivity and tortuosity improves the design.

Chapter 7

Results and Discussion

7.1 Overview

[What this chapter contains.]

7.2 Helmholtz Solver Validation

7.2.1 Free-Air Propagation

[Compare numerical wave speed and phase against analytical solution.]

7.2.2 Hard-Wall Cavity Modes

[Compare resonant frequencies against analytical 2D cavity modes.]

7.2.3 Uniform Porous Slab

[Compare transmission/reflection against transfer-matrix reference for a uniform JCA slab.]

7.3 Design-Region Demonstrations

7.3.1 Task Definition

[Describe the chosen wave-computing task: e.g. two-frequency routing, logic gate, or binary classification.]

7.3.2 Baseline Designs

[Results for homogeneous or simple periodic designs.]

7.3.3 Optimised Designs

[Convergence plots, final block patterns, field snapshots, performance metrics.]

7.4 Design-Space Ablations

[Compare ϕ -only, $\phi + \sigma$, and $\phi + \sigma + \alpha_\infty$ designs in terms of final loss, pattern complexity, and physical interpretability.]

7.5 Discussion

[Interpret results, limitations, relation to literature.]

Chapter 8

Conclusions and Future Work

8.1 Summary of Contributions

[Bullet summary of what was achieved.]

8.2 Limitations

- Frequency-domain formulation assumes linear time-invariant steady-state response; transient and nonlinear effects are not captured.
- Gradient-free optimisation limits design-space size and convergence speed.
- 2D approximation; real porous devices are three-dimensional.
- Fixed Λ and Λ' reduce the physical realism of the design space.
- Mapping from optimised effective properties to a manufacturable geometry is not addressed.

8.3 Future Work

1. Extend to time-domain FDTD or auxiliary differential equations for broadband transient signals.
2. Implement adjoint-based gradient optimisation for larger design regions.
3. Relax fixed Λ/Λ' or link them to a pore-scale geometric model.
4. Validate select designs in OpenFOAM or experimentally.
5. Demonstrate a concrete classification or routing task with quantified accuracy and generalisation.

8.4 Concluding Remarks

[Final paragraph.]

References

- [1] Biot MA. Theory of Propagation of Elastic Waves in a Fluid-Saturated Porous Solid. I. Low-Frequency Range. The Journal of the Acoustical Society of America. 1956;28(2):168-78.
- [2] Lin X, Rivenson Y, Yardimci NT, Veli M, Luo Y, Jarrahi M, et al. All-Optical Machine Learning Using Diffractive Deep Neural Networks. Science. 2018;361(6406):1004-8.
- [3] Stern M, Arinze C, Perez L, Palmer SE, Murugan A. Supervised Learning Through Physical Changes in a Mechanical System. Proceedings of the National Academy of Sciences. 2020;117(26):14843-50.
- [4] Stern M, Hexner D, Rocks JW, Liu AJ. Supervised Learning in Physical Networks: From Machine Learning to Learning Machines. Physical Review X. 2021;11(2):021045.
- [5] Wright LG, Onodera T, Stein MM, Wang T, Schachter DT, Hu Z, et al. Deep Physical Neural Networks Trained with Backpropagation. Nature. 2022;601(7894):549-55.
- [6] Momeni A, Rahmani B, Malléjac M, del Hougne P, Fleury R. Backpropagation-Free Training of Deep Physical Neural Networks. Science. 2023;382(6676):1297-303.
- [7] Hughes TW, Williamson IAD, Minkov M, Fan S. Wave Physics as an Analog Recurrent Neural Network. Science Advances. 2019;5(12):eaay6946.
- [8] Silva A, Monticone F, Castaldi G, Galdi V, Alù A, Engheta N. Performing Mathematical Operations with Metamaterials. Science. 2014;343(6167):160-3.
- [9] Zangeneh-Nejad F, Sounas DL, Alù A, Fleury R. Analogue Computing with Metamaterials. Nature Reviews Materials. 2021;6(3):207-25.
- [10] Zuo S, Wei Q, Tian Y, Cheng Y, Liu X. Acoustic Analog Computing System Based on Labyrinthine Metasurfaces. Scientific Reports. 2018;8:10103.
- [11] Zhang T, Cheng Y, Guo JZ, Xu JY, Liu XJ. Acoustic Logic Gates and Boolean Operation Based on Self-Collimating Acoustic Beams. Applied Physics Letters. 2015;106(11):113503.

- [12] Zhang T, Cheng Y, Yuan BG, Guo JZ, Liu XJ. Compact Transformable Acoustic Logic Gates for Broadband Complex Boolean Operations Based on Density-Near-Zero Metamaterials. *Applied Physics Letters*. 2016;108(18):183508.
- [13] Wang Y, Xia JP, Sun HX, Yuan SQ, Liu XJ. Binary-phase acoustic passive logic gates. *Scientific Reports*. 2019;9(1):8355.
- [14] Xia YH, Pang NQ, Wang Y, Wang LX, Ge Y. Ultra-broadband acoustic logic gate based on passive phase manipulation. *Chinese Physics B*. 2026;35(1):014301.
- [15] Parsa A, Wang D, O'Hern CS, Shattuck MD, Kramer-Bottiglio R, Bongard J. Evolution of Acoustic Logic Gates in Granular Metamaterials. In: *International Conference on the Applications of Evolutionary Computation (Part of EvoStar)*. Cham: Springer; 2022. p. 93-109.
- [16] Botteldooren D. Finite-Difference Time-Domain Simulation of Low-Frequency Room Acoustic Problems. *The Journal of the Acoustical Society of America*. 1995;98(6):3302-8.
- [17] Bilbao S. Numerical Sound Synthesis: Finite Difference Schemes and Simulation in Musical Acoustics. Chichester, UK: John Wiley & Sons; 2009.
- [18] Taflov A, Hagness SC. Computational Electrodynamics: The Finite-Difference Time-Domain Method. 3rd ed. Boston, MA: Artech House; 2005.
- [19] Yee KS. Numerical Solution of Initial Boundary Value Problems Involving Maxwell's Equations in Isotropic Media. *IEEE Transactions on Antennas and Propagation*. 1966;14(3):302-7.
- [20] Zwikker C, Kosten CW. Sound Absorbing Materials. New York: Elsevier; 1949.
- [21] Johnson DL, Koplik J, Dashen R. Theory of Dynamic Permeability and Tortuosity in Fluid-Saturated Porous Media. *Journal of Fluid Mechanics*. 1987;176:379-402.
- [22] Champoux Y, Allard JF. Dynamic Tortuosity and Bulk Modulus in Air-Saturated Porous Media. *Journal of Applied Physics*. 1991;70(4):1975-9.
- [23] Allard JF, Atalla N. Propagation of Sound in Porous Media: Modelling Sound Absorbing Materials. 2nd ed. Chichester: John Wiley and Sons; 2009.
- [24] Biot MA. Nonlinear and Semilinear Rheology of Porous Solids. *Journal of Geophysical Research*. 1973;78(23):4924-37.
- [25] Jaafar AH, Shao L, Dai P, Zhang T, Han Y, Beanland R, et al. 3D-Structured Mesoporous Silica Memristors for Neuromorphic Switching and Reservoir Computing. *Nanoscale*. 2022;14(46):17170-81.
- [26] Ding G, Zhao J, Zhou K, Zheng Q, Han ST, Peng X, et al. Porous Crystalline Materials for Memories and Neuromorphic Computing Systems. *Chemical Society Reviews*. 2023;52(20):7071-136.

-
- [27] Adamatzky A. A Brief History of Liquid Computers. *Philosophical Transactions of the Royal Society B: Biological Sciences*. 2019;374(1774):20180372.
- [28] Baltussen MGJ, Dorel R, Hendriksen DE, Pfeifer T, van Bodegraven HJ, Huck WTS, et al. Chemical Reservoir Computation in a Self-Organizing Reaction Network. *Nature*. 2024;633(8028):87-95.
- [29] Prakash M, Gershenfeld N. Microfluidic Bubble Logic. *Science*. 2007;315(5813):832-5.
- [30] Boulvert J, Cavalieri T, Costa-Baptista J, Schwan L, Romero-García V, Gabard G, et al. Optimally Graded Porous Material for Broadband Perfect Absorption of Sound. *Journal of Applied Physics*. 2019;126(17):175101.
- [31] Pan B, Song X, Xu J, Sui D, Xiao H, Zhou J, et al. Accelerated Inverse Design of Customizable Acoustic Metaporous Structures Using a CNN-GA-Based Hybrid Optimization Framework. *Applied Acoustics*. 2023;210:109445.
- [32] Bérenger JP. A Perfectly Matched Layer for the Absorption of Electromagnetic Waves. *Journal of Computational Physics*. 1994;114(2):185-200.

Appendix A

Additional Derivations

[Insert supplementary derivations, proofs, or data here.]